



Zoology Legends

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Principle of PCR and its reagents.

Principle of PCR

The principle of PCR (Polymerase Chain Reaction) involves the enzymatic amplification of a specific DNA sequence. It consists of repeated cycles of heating (denaturation), cooling (annealing), and DNA synthesis (extension). This process uses a DNA template, DNA primers, and a heat-stable DNA polymerase to exponentially produce copies of the target DNA region, facilitating various genetic analyses.

Reagent use in PCR.

Some of the key reagents used in a typical PCR reaction:

1. DNA Template: The starting DNA strand that contains the target sequence you want to amplify.
2. Primers: Short DNA sequences that define the start and end points of the target region to be amplified.
3. DNA Polymerase: The enzyme responsible for synthesizing a complementary DNA strand using the primers and nucleotides.
4. Nucleotides (dNTPs): The building blocks of DNA strands, containing adenine (A), thymine (T), cytosine (C), and guanine (G).
5. Buffer Solution: A solution that provides the optimal pH and salt concentration for the PCR reaction to occur.



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PCR uses /Applications of PCR

PCR has applications in DNA amplification, Genetic testing, forensics, medical diagnostics, Environmental studies, Evolutionary studies, food safety, Archaeology, and anthropology

Which diseases are caused by x-linked disorder?

X-linked recessive diseases are genetic disorders caused by mutations in genes located on the X chromosome. Because females have two X chromosomes, they can be carriers of X-linked recessive mutations without showing symptoms. Males, who only have one X chromosome, are more likely to exhibit symptoms of X-linked recessive diseases if they inherit a mutant allele. Examples of X-linked recessive diseases include hemophilia, Duchenne muscular dystrophy, Huntington's Disease, and red-green color blindness.

Difference b/w genetic & environmental causes of variations

Variations in traits can be caused by both environmental and genetic factors. Environmental factors include things like diet, exposure to toxins, and physical activity, which can influence gene expression and affect traits. Genetic factors, on the other hand, involve variations in DNA sequences that are inherited from parents and can directly impact traits. It's important to note that in some cases variations are a result of a combination of both environmental and genetic influences.



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Erythrocyte lysis In RNA/Erythrocyte extraction from RNA purification

Erythrocyte lysis is a process used to break open red blood cells (erythrocytes) in order to release their contents, including RNA. This process is commonly used in RNA extraction protocols to obtain RNA samples from whole blood, which contains a mixture of various cell types, including erythrocytes, leukocytes, and platelets.

One commonly used method for erythrocyte lysis in RNA extraction is the use of a hypotonic buffer, such as ammonium chloride, which causes the erythrocytes to swell and burst, releasing their contents into the solution. The leukocytes and platelets are relatively resistant to lysis under these conditions, allowing for the isolation of RNA from these cell types.

After erythrocyte lysis, the RNA can be extracted from the solution using various methods, such as column-based or bead-based purification. The extracted RNA can then be further purified and quantified for downstream applications, such as gene expression analysis or sequencing.

Procedure of western blotting

The Western blotting procedure involves several (6) steps. First, proteins are separated by size using gel electrophoresis. Then, the proteins are transferred onto a membrane. The membrane is blocked to prevent non-specific binding, and then incubated with a primary antibody that binds to the target protein. After washing, a



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secondary antibody is added that recognizes the primary antibody. Finally, the protein bands are visualized using a detection method such as chemiluminescence.

Hot start and suicidal PCR

Hot start PCR is a technique used to prevent non-specific amplification during the initial steps of the polymerase chain reaction (PCR). This is achieved by adding a DNA polymerase inhibitor to the reaction mix that becomes inactive at lower temperatures. This helps to reduce background noise and improve specificity in PCR reactions.

Hot start PCR is a modification of standard PCR in which the reaction mix is heated to an intermediate temperature (usually 95-98°C) before the addition of the Taq polymerase enzyme. This allows the reaction to undergo an initial denaturation step without the presence of the polymerase, which can reduce nonspecific amplification caused by the presence of primer-dimer or mis-primed products. After the initial denaturation, the reaction is cooled to the optimal temperature for Taq polymerase activity and the polymerase is added. This technique can improve specificity and sensitivity of PCR, especially for difficult targets or complex templates.

Suicide PCR, on the other hand, is a technique that uses a modified primer in which the 3' end of the primer contains a degenerate base that can only be extended by the correct target DNA template. If the target DNA is not present, the primer will not be extended and will act as an inhibitor for subsequent PCR reactions. This helps to reduce the formation of nonspecific amplification products and can improve the specificity of PCR. The modified primer is designed to be



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used in a limited-cycle PCR reaction, in which a small number of cycles (e.g., 1-3 cycles) are performed before the reaction is stopped. The products of the limited-cycle PCR are then used as a template for a standard PCR reaction, which is carried out with regular primers to amplify the target DNA specifically. This technique is particularly useful for amplifying low-abundance targets or targets that are difficult to amplify with standard PCR.

Suicide PCR method prescribes the use of any primer combination only once in a PCR (hence the term “suicide”), which should never have been used in any positive control PCR reaction, And the primers should always target a genomic region never amplified before in the lab using this or any other set of primers

Type/Pattern of genetic disorders?

Patterns of genetic disorders refer to recognizable ways in which genetic conditions are inherited within families. These include:

1. **AUTOSOMAL DOMINANT**: Affected individuals are heterozygote. One allele is mutated, while other one is normal.
2. **AUTOSOMAL RECESSIVE**: Homozygous, when both alleles of the gene are mutated.
3. **X-LINKED RECESSIVE**: Mutated alleles are present on either of the sex chromosomes.
4. **MULTIFACTORIAL**: Genes and environment both contribute.



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Ploidy and its types?

“Multiples of haploid number”

- Triploidy or tetraploidy etc
- Fetus does not survive

Two major types

- Autopolyploidy
- Allopolyploidy

Types of ploidy

- ☐ Triploid (3x) watermelon
- ☐ Tetraploid (4x) - cotton
- ☐ Pentaploid (5x) - Kenai Birch
- ☐ Hexaploid (6x) - wheat

Causes of ploidy

- ☐ Retention of polar body.
- ☐ Formation of diploid sperms.
- ☐ Di-spermy – fertilization by two sperms.



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Polyploidy in Plants

- ☐ **Triploid:** apple, banana, citrus, ginger, watermelon
- ☐ **Tetraploid:** cotton, potato, tobacco, peanut
- ☐ **Hexaploid:** oat, kiwifruit and wheat
- ☐ **Octaploid:** strawberry
- ☐ **Decaploid:** some sugar cane hybrids

Prophase II of meiosis?

During Prophase II of meiosis II:

- ☐ No replication of Chromosomes as they are already replicated. Centrioles are separated and take positions on the opposite poles of the cell
- ☐ Nuclear membrane disappears.
- ☐ Spindle fibers form and radiate toward the center of the cell.
- ☐ Chromosomes align in the center of the cell.

VNTR and its applications.

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes, Show variations in length between individuals.

Application:

- DNA fingerprinting



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- Genetic diversity
- Forensic mapping of genome
- breeding pattern of wild and domesticated animals
- Paternity testing
- Evolutionary research.

Continuous and discontinuous variations.

Discontinuous Variation

- Discontinuous variations are entirely genetically controlled.
- They cannot be altered by external conditions. Blood group, color blind, dwarfism are genetically controlled.

Continuous variation

- ☐ Continuous variation occurs –when controlled by the genes and the environment.
- ☐ Height depend on the genes and on the amount of food you eat during your growing.

Trisomy and its types?

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two. Trisomy is a type of aneuploidy (an abnormal number of chromosomes).



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Trisomy -Autosomes and sex chromosome

- Autosomal trisomy
- Sex-chromosome trisomy
- Trisomy of chromosome 21, which is found in Down syndrome, is called trisomy 21.

Trisomy Types

- **Primary trisomy** - an entire extra chromosome has been copied.
- **Partial trisomy** - an extra copy of part of a chromosome.
- **Secondary trisomy** - the extra chromosome has duplicated arms.
- **Tertiary trisomy** - extra chromosome is made up of copies of arms from two other chromosomes.

Common trisomy in humans

- Trisomy 21
- Trisomy 18
- Trisomy 13
- Klinefelter's syndrome (47 XXY)



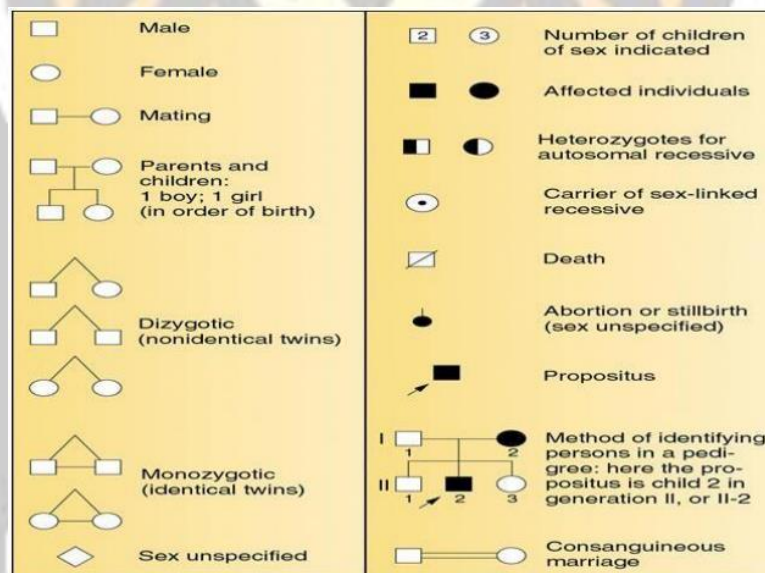
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Genetics mapping causes disease

The use of molecular markers to identify the location of genes on a chromosome is called Genetic Mapping.

- Genetic mapping - also called linkage mapping provides clues about which chromosome contains the gene and precisely where it lies on that chromosome.
- Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

Pedigree diagram



Define leading strand, lagging strand, Okazaki fragments, Gyrase and sister chromatids

In DNA replication, the leading strand is synthesized continuously in the 5' to 3' direction, following the replication fork, while the lagging strand is synthesized



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discontinuously in short fragments called Okazaki fragments. Okazaki fragments are later joined by DNA ligase to form a continuous strand. Gyrase is an enzyme that helps relieve the tension caused by DNA unwinding during replication. Sister chromatids are two identical copies of a chromosome joined together at the centromere, formed during DNA replication, and are separated during cell division.

Difference between Ring and Philadelphia chromosome:

A **ring chromosome** is a chromosome whose arms have fused together to form a ring. In humans it is denoted by small r and in drosophila it is denoted by capital R.

Philadelphia chromosome:- is a specific chromosomal abnormality that is related with chronic myelogenous leukemia(CML). Reciprocal translocation between chromosome 9 and chromosome 22 which is specifically designated.

Autosomal recessive Disease

An autosomal recessive disease is a genetic disorder caused by inheriting two copies of a mutated gene, one from each parent, who are typically carriers of the mutated gene. Examples of autosomal recessive diseases include cystic fibrosis, sickle cell anemia, and Tay-Sachs disease. These conditions often appear when an individual inherits two copies of the faulty gene, leading to the expression of the disease's symptoms. Carriers of a single copy of the mutated gene usually do not show symptoms but can pass the gene on to their offspring.

Structural abnormalities in Chromosomes

1. Deletion



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2. Duplication
3. Translocation
4. Inversion
5. Ring Chromosome 6. Isochromosome

Geimsa stain

Geimsa stain is a type of stain used in microscopy to visualize and differentiate different types of cells, especially blood cells. It works by staining the DNA and RNA in the cells, allowing for better visualization of cellular structures and identification of specific cell types.

DNA polymorphism

DNA polymorphism is a sequence difference compared to a reference standard present in at least 1–2% of a population. DNA Polymorphism may be single nucleotide or thousands of BASES.

Polymorphisms can be due to a single base or thousands of bases. DNA Polymorphism may or may not have phenotypic effects. DNA Polymorphism are found throughout the genomes

RAPD & ALFP difference

RAPD: A PCR-based technique used to amplify random segments of genomic DNA using single primers. The amplified products reveal polymorphisms between individuals. Generally less expensive and simpler, as it requires fewer primer pairs.



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AFLP: A PCR-based method that selectively amplifies subsets of restriction enzyme-digested genomic DNA, creating a fingerprint of fragments that differ among individuals. Usually more expensive and complex, involving multiple primer pairs and enzyme digestion steps.

Organic and inorganic method of DNA extraction:

Organic Method: The organic method of DNA extraction involves cell disruption using physical methods like grinding or enzymatic digestion. Proteins are removed with detergents and enzymes, followed by DNA precipitation using organic solvents like ethanol. After washing, purified DNA is eluted in a buffer for further analysis.

Inorganic Method: The inorganic method employs silica columns or magnetic beads coated with silica-binding materials. DNA binds to these surfaces, while contaminants are washed away. Elution releases pure DNA, suitable for downstream applications. This method minimizes organic solvent use and is amenable to automation

Epistasis and its types

Alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

Example: Epistasis in horses fur

- Other genes can add effects to base colors.
- Horse coloration involves 2 or more gene pairings.



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- EE or Ee is for black.
- ee is for red (sorrel).

Effects of point mutation.

Point mutations alter single DNA bases, affecting protein structure/function.

Substitutions (missense, nonsense) change amino acids or stop codons, impacting gene expression and possibly causing disease or genetic diversity.

Properties of autosomal recessive inheritance

- **Consanguinity:** These diseases appear where parents have common ancestors. Same allele increases chances of inheriting disorders: The chance of inheriting identical alleles by an individual from both parents increases the chance of inheriting a recessive disorder.
- **Common genetic background:** Small populations of individuals with a common genetic background may have increased risk of recessive diseases.
- **Ashkenazi:** Gaucher disease
- Phenotype found more likely in siblings of probed than in other relatives. Males and females are equally affected. In most of the cases, parents of affected individuals are asymptomatic.
- Risk for each of the sibling being affected.
- Risk of disease for each sib is 25 %

Briefly describe the organic method of DNA extraction.

1. Cell Disruption: Cells are broken open to release DNA.



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2. Protein Removal: Enzymes and detergents eliminate proteins.
3. DNA Precipitation: Ethanol or isopropanol is added to cause DNA to precipitate.
4. Washing: DNA is washed and purified.
5. Elution: DNA is dissolved in a buffer for use in analysis or experimentation.

Write a note on non-disjunction.

Non-disjunction is a genetic event that can occur during cell division, specifically during the separation of chromosomes. It can be caused by various factors, including genetic abnormalities, environmental factors, and maternal age. Non-disjunction can lead to an abnormal number of chromosomes in the resulting cells, which can cause genetic disorders such as Down syndrome.

Write the dominant and recessive alleles for eye color, hair, facial features, and appendages.

Here's a breakdown of which traits are typically considered dominant and recessive:

1. Eye Color

- Dominant: Brown
- Recessive: Blue

2. Hair Color:

- Dominant: Dark colors (Black, Brown)



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- Recessive: Light colors (Blonde, Red)

3. Facial Features (Dimples as an example):

- Dominant: Dimples

- Recessive: No dimples

4. Appendages (Freckles on arms as an example):

- Dominant: Freckles on arms

- Recessive: No freckles on arms

Write all procedures of staining method

Staining procedures have been developed in the past two decades and these techniques help to study the karyotype in plants and animals.

Feulgen staining

- ☐ Cells are subjected to a mild hydrolysis in 1N HCl at 60 °C for 10 minutes.
- ☐ This treatment produces a free aldehyde group in deoxyribose molecules.

Q Banding

- ☐ Q bands are the fluorescent bands observed after quinacrine mustard staining with UV light.
- ☐ The distal ends not stained by this technique.



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R Banding

The R bands (reverse) are those located in the zones that do not fluoresce with the quinacrine mustard, that is they are between the Q bands and can be visualized as green.

C Banding:

The C bands correspond to constitutive heterochromatin. The heterochromatin regions in a chromosome distinctly differ in their stainability from euchromatic region

Write about short tandem repeats polymorphism.

- ☐ STR is simple repeats of nucleotide sequences.
- ☐ Mononucleotide, dinucleotide
- ☐ Used in linkage analysis Used in DNA forensics.

Different between the wild types of phenotype and genotypes Mutations, Forward, Reverse and Neutral Mutations

The most common phenotype in natural populations of the organism is called as wild type phenotype. The effect of mutation is considered with reference to wild type phenotype.

Forward mutation: Mutation that alters the wild type phenotype into mutant phenotype.

Reverse mutation: Mutation that changes a mutant phenotype back into the wild type.



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Neutral mutation: Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected AA has little influence on protein function.

Co-dominance with 3 examples

- ☐ A situation, when both alleles appear in the phenotype.
- ☐ Neither allele is dominant.
- ☐ Both alleles are expressed in a heterozygous individuals.
- ☐ Parallel behavior of both allele.

Example 1 : Blood group

Example 2: Roan cattle inheritance

- ☐ Roan color in cattle appears as a mix of red and white colors.

crcr = red hairs cwcw = white hairs

crcw = roan coat

Example 3: Appaloosa horse

- ☐ Gray horse dominant to white horse.
- ☐ Heterozygous is appaloosa –a white horse with gray spots.

Linked and causative single linked polymorphism

A ***linked single nucleotide polymorphism (SNP)*** is located near another genetic variant on the same chromosome, often inherited together. A **causative SNP**



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directly influences a trait or disease, demonstrating a functional relationship through its impact on gene expression or protein structure.

Write names of Steps of epigenetic Modifications

Followings are the steps of epigenetics modifications

- Cytosine methylation
- Histones modification
- Non-coding RNAs (siRNAs)
- All above regulate gene expression

Which factors use to make pedigree diagram

Generations of a pedigree as roman numerals: Generations in a pedigree diagram are numbered, by using Roman numerals, starting with the parental generation, at the top of the diagram as generation I. Members of each generation are numbers from left to right. For convenience, the members of each generation are numbered across the line, from left to right, using normal numerals.

Define penetrance

Percent individuals with a given allele that show the phenotype of the allele

Proportion of individuals carrying a particular variant of a gene (allele or genotype) that also expresses an associated trait (phenotype). Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.



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Enlisted the reagents used in RNA purification. How these reagents contribute to RNA purification

1. Collect 250 uL samples to a 1.5 mL eppendorff and add 750uL Trizol and samples is overtaxed for 15 sec. incubate at room temperature for 7 minutes.
2. Pulse spin to remove liquid from the tube lid.
3. Add 200uL 100% chloroform to the sample, vortex for 15 sec and incubate at room temperature for 7 minutes.
4. Centrifuge at 12,000g for minutes at room temperature.
5. Transfer 450uL of the upper aqueous layer to a separate microcentrifuge tube.
6. Add 500uL 100% isopropanol, invert tube several time to mix and hold at room temperature for 10 minutes.
7. Discard supernatant. Care should be taken to assure that the RNA pellet is not disturbed.
8. Add 1 mL of 80% ethanol. Mix gently.
9. Centrifuge at 10,000g for 5 minute at 4C.
10. Discard ethanol. Invert tube on a clean tissue wipe and allow to air dry for 10 minutes. It is important not to let the RNA pellet over dry as this will decrease its solubility.
11. Hydrate pellet in 50 uL of RNase free water and store at -40 to -80 °C.



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Write about variation in autosomal dominant inheritance in reference of following terms: Penetrance, expressivity, pleiotropy.

Penetrance

Percent individuals with a given allele that show the phenotype of the allele

Proportion of individuals carrying a particular variant of a gene (allele or genotype) that also expresses an associated trait (phenotype). Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.

Expressivity

Extent to which allele is expressed at phenotypic level affected by genetic background and environment. Difference likelihood of the gene generating its associated phenotype. Influence of an expressed gene at the level of particular individuals.

Pleiotropy

The term is analogous to the severity of a condition in clinical medicine. Variable expressivity when phenotype is expressed to a different degree among individuals with the same genotype.

Types of chromosomes

There are two types of eukaryotic chromosomes:

➤ Autosomes



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➤ Sex chromosomes

Autosomes: Paired chromosomes with the same length, shape, centromere location, and genes. Any chromosomes other than sex chromosomes are called as autosomes.

Sex Chromosomes: Members of a pair of chromosomes that differ between males and females. Based on this pair of chromosomes, sex is identified.

Type of Chromosomes based on Centromere position

Chromosomes may differ in the position of the centromere.

Centromere is near the middle, **metacentric**

Centromere is toward one end, **acrocentric** or sub metacentric

Centromere is very near to end, **telocentric**

What is full penetrant and low penetrant?

- ☐ Fully penetrant, where other genes and environmental factors have no effect.
- ☐ Low-penetrance, genes play a small part, along with other environmental factors.

Genetic/inherited variation and its two examples

Genetic or inherited variations refer to the differences in genetic information that are passed down from parents to their offspring. These variations can occur due to changes in DNA sequences, such as mutations or genetic recombination during the formation of reproductive cells.



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Some examples of genetic variations include:

1. Eye color
2. Blood type
3. Height
4. Skin color
5. Inherited diseases

Frequency of genetic disorders

Frequency of autosomal dominant disorders: Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals.

Frequency of autosomal recessive disorders: Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

Frequency of x-linked disorders: Frequency of X-linked disorders is between 0.5 – 2 per 1000 individuals.

Frequency of chromosomal disorders: Frequency of chromosomal disorders is between 6-9 per 1000 individuals.

Frequency of multifactorial disorders: Frequency of multifactorial disorders is between 20 – 50 per 1000 individuals.

Most common multifactorial disorders are cancers, diabetes, heart diseases.

What is mutagens?

Mutagens are agents that increase the rate of mutation above the spontaneous mutation rate. They can induce changes in the DNA sequence by causing chemical modifications, DNA strand breaks, or other forms of damage. Mutagens can be classified into two main categories:

Physical Mutagens: These mutagens include physical agents that directly interact with DNA and cause damage. Examples of physical mutagens include ionizing



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radiation (such as X-rays and gamma rays) and ultraviolet (UV) radiation. These agents can lead to the formation of DNA lesions and errors during DNA repair.

Chemical Mutagens: Chemical mutagens are compounds that interact with DNA and cause chemical modifications. These modifications can lead to base substitutions, insertions, deletions, or other structural changes in the DNA sequence. Examples of chemical mutagens include certain chemicals found in the environment, tobacco smoke, industrial pollutants, and some cancer-causing agents.

Lethal mutation

Causes the death of the organism. Death does not have to occur immediately, it may take several months or even years. Longevity of an individual is significantly reduced.

Applications of molecular markers – short range

- Parentage determination
- Genetic distance estimation
- Determination of twin zygosity
- Sexing of pre-implanted embryos
- Identification of disease carriers

Applications of Molecular Markers – Long Range

- Gene mapping & mapping of QTL by linkage
- Markers assisted selection of breeds



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Translocations

Transfer of genetic material from one chromosome to another. There are **two types** of translocations

- Reciprocal translocation
- Robertsonian translocation

Balanced translocations where exchange of no extra genetic information or missing genes. Unbalanced translocations where exchange of genetic information is extra or missing some of the genes.

Reciprocal an exchange of material between two different chromosomes is called a reciprocal translocation. One in each of 600 births in humans.

Robertsonian Translocations

Breaks occur at the extreme ends of the short arms of two non-homologous acrocentric chromosomes. The small acentric fragments are lost. The larger fragments fuse at their centromere regions to form a single chromosome

The most common translocation in humans involves chromosomes 13 and 14 and is seen in about 0.97 / 1000 newborns.



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Common human diseases caused by translocation

- Cancer
- Infertility
- Down syndrome
- Leukemia
- XX male syndrome

RNA extraction by Organic Method

Step 1: Lyse/homogenize cells using trizol reagent/phenol chloroform and centrifuge.

- ☐ Step 2: Organic phase separates from aqueous phase organic solvents at bottom.
- ☐ Step 3: Aqueous phase on top (contains total RNA).

Step 4: Cellular debris and genomic DNA appears as a “film” of debris at the interface of the two solutions.

- ☐ Step 5: Remove RNA solution to a clean tube.
- ☐ Step 6: Precipitate RNA and wash with ethanol, then resuspend RNA in water



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SOUTHERN BLOTTING PROCEDURE.

1. DNA Digestion: The DNA sample is cut into fragments using restriction enzymes.
2. Gel Electrophoresis: Fragments are separated by size using gel electrophoresis.
3. Denaturation: DNA fragments are treated to separate the double-stranded DNA into single strands.
4. Transfer: Fragments are transferred from the gel to a membrane, like nitrocellulose or nylon.
5. Hybridization: A labeled DNA probe binds to the target DNA sequences on the membrane.
6. Washing: Non-specifically bound probes are washed away, leaving only the hybridized ones.
7. Detection: The presence of the labeled probe is visualized using autoradiography or other detection methods.

This technique is used to detect specific DNA sequences and analyze DNA fragment sizes

RBC lysis in RNA extraction:

Red Blood Cell (RBC) lysis is a crucial step in RNA extraction from blood samples. It involves breaking down RBCs to release hemoglobin, preventing interference with RNA analysis. Lysis buffers containing detergents and salts are used to rupture RBC membranes. This step ensures the isolation of high-quality



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RNA from white blood cells, which are then processed for downstream applications like gene expression analysis.

Explain the haploid-diploid sex determination system

In the haploid-diploid sex determination system, males develop from unfertilized eggs (haploid, containing half the usual genetic material) and are thus haploid. Females develop from fertilized eggs (diploid, with the full genetic material), making them diploid. This system is observed in certain insects, like bees and ants. It results in a sex ratio of 1:1 as females inherit half their genetic material from their fathers (haploid) and the other half from their mothers (diploid).

Multifactorial diseases

Genetic disorders: A genetic disease or disorder appear as a result of mutations in an individual's DNA. Change in DNA sequence may be small or large like deletions/addition of entire chromosome.

Multifactorial diseases: Genetic disorders controlled by many genes plus the effects of the environment. Multifactorial diseases • Asthma • Cancers • Ciliopathies • Cleft palate • Diabetes • Heart diseases • hypertension • inflammatory bowel disease • intellectual disability • mood disorder • Obesity • infertility

Mendelian and non Mendelian

Mendelian genetics: Mendelian genetics - genes that obey Mendel's laws. Most genes follow a Mendelian pattern of inheritance, However, there are many that don't.



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Non-Mendelian genetics: Patterns of inheritance that deviate from a Mendelian pattern. Linkage -Non-Mendelian. Maternal effect, epigenetic inheritance, Mitochondrial inheritance etc

Metaphase II of Meiosis

Metaphase II of meiosis In metaphase II, sister chromatids are arranged at the metaphase plate. The kinetochores of sister chromatids attach to microtubules extending from opposite poles.

Chromosomes lined up across the middle of the cell. Due to crossing over during meiosis I, the two sister chromatids of each chromosome are no longer genetically identical.

How to write mutations in early exon region?

Mutation in intronic sequence

Beginning of intron

□ The number of the last nucleotide of the preceding exon, a plus sign and the position in the intron, like c.88+2T>G

End of intron

□ The number of the first nucleotide of the following exon, a minus sign and the position upstream in the intron, like c.89-1G>T



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Difference of mitosis and meiosis

Feature	Mitosis	Meiosis
Number of Divisions	One division	Two divisions (meiosis I and II)
Resulting Cells	Produces two identical diploid cells	Produces four genetically diverse haploid cells
Chromosome Number	Maintains the same chromosome number	Reduces chromosome number by half
Genetic Variation	Limited genetic variation	Significant genetic variation through crossing over and independent assortment
Occurrence	Occurs in somatic cells	Occurs in reproductive cells
Purpose	Cell replication, growth, and tissue repair	Production of gametes for sexual reproduction

How gene product predict that gene did not working properly?

Predicting that a gene product won't do the job

- ☐ Deletion, nonsense, frameshift of sequence is deleterious.



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- ☐ Mutation in splice site usually has severe effects.
- ☐ Missense mutations depend on location in protein.
- ☐ Is it non-conservative
- ☐ Is AA conserved in evolution

Process of cytokinesis in plants?

Cytokinesis is the division of the cytoplasm into two individual cells.

Process differs in plants and animal cells

Cytokinesis in animal cells: Cell membrane forms a cleavage furrow that eventually pinches the cell into two equal parts. Each part containing its own nucleus and cytoplasmic organelles.

Cytokinesis in Plants cells: a cell plate is formed in midway that divides the nuclei. Cell plate gradually develops into a separating membrane.

Write 3 types of genetic disorder.

Down syndrome: A chromosomal disorder caused by an extra copy of chromosome 21, leading to cognitive and physical developmental delays.

Cystic Fibrosis: A hereditary disorder affecting the lungs and digestive system, characterized by the production of thick and sticky mucus that can obstruct airways and lead to respiratory and digestive problems.



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Huntington's disease: An inherited neurological disorder that causes progressive degeneration of brain cells, leading to motor dysfunction, cognitive decline, and emotional disturbances.

What is addition mutation?

Addition mutation actually results in a frame shift mutation. Often, it has severe effects on protein function. It is also called as insertions.

Prophase I of meiosis

Prophase I occupies more than 90% of the time required for meiosis.

Five sub stages of prophase I

Sub stages of prophase I

- ☐ Leptotene
- ☐ Zygotene
- ☐ Pachytene
- ☐ Diplotene
- ☐ Diakinesis

Events of prophase I: The duplicated homologous chromosomes pair. The nucleolus disappears. Centrioles migrate to opposite poles of the cell. Meiotic spindle forms between the two pairs of centrioles. Nuclear membrane disappears.



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Chiasmata formation during prophase I During pairing up of chromosomes, nonsister chromatids exchange DNA segments. Each pair of chromosomes forms a tetrad, a group of four chromatids.

HGCN rules

HGCN rule for substitution, range of residue duplication deletion insertion inversion. The HGCN rule, or the Human Genome Variation Society nomenclature, provides guidelines for describing genetic variations in DNA, RNA, and protein sequences. The HGCN rule includes specific conventions for representing different types of variations, such as substitutions, insertions, deletions, and rearrangements, in a standardized format. This allows researchers and clinicians to accurately communicate and interpret genetic variations across different studies and databases. The HGCN rule plays a crucial role in facilitating the understanding and sharing of genetic information in the scientific community.

- ☐ "c." for a coding DNA sequence (like c.76A>T)
- ☐ "g." for a genomic sequence (like g.476A>T)
- ☐ "m." for a mitochondrial sequence (like m.8993T>C)
- ☐ "r." for an RNA sequence (like r.76a>u)
- ☐ "p." for a protein sequence (like p.Lys76Asn)

Diameter and length of diptera chromosome

The giant chromosomes found in diptera and they may be as long as 300 μ and up to 10 μ in diameter



Zoology Legends

Blending and particulate hypothesis

Blending hypothesis

- ☐ According to blending hypothesis, genetic material of the parents mixes during fertilization.
- ☐ Blue and yellow paints blend to make green paint.

Particulate hypothesis

- ☐ According to particulate hypothesis, parents pass on distinct heritable units.
- ☐ Those distinct heritable units are now called as genes.

Symbols of insertion, duplication, inversion, deletion mutations

Specific changes in sequences are

- ☐ ">" indicates a substitution at DNA level (like c.76A>T)
- ☐ "_" (underscore) indicates a range of affected residues, separating the first and last residue affected (like c.76_78delACT).
- ☐ "del" indicates a deletion (like c.76delA).
- ☐ "dup" indicates a duplication (like c.76dupA); duplicating insertions are described as duplications, not as insertions.
- ☐ "ins" indicates an insertion (like c.76_77insG)
- ☐ "inv" indicates an inversion (like c.76_83inv)



Zoology Legends

Name of Polymorphic types of DNA sequences.

- ☐ RFLP: Restriction Fragment Length Polymorphism
- ☐ VNTR: Variable Number Tandem Repeats (8 to > 50 base pairs)
- ☐ STR: Short Tandem Repeats (1– 8 base pairs).
- ☐ SNP: Single Nucleotide Polymorphisms
- ☐ AFLP: Amplified Fragment Length Polymorphism

Five basic complications in the pattern of inheritance.

Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance

- ☐ A common recessive inheritance can give dominant pattern
- ☐ Autosomal dominant inheritance with non-penetrance
- ☐ Autosomal dominant inheritance with variable expression.

Heredity and particulate hypothesis

Heredity

- ☐ Heredity is transmission of traits from parents to offspring.
- ☐ Transmission is particulate.



Zoology Legends

Particulate hypothesis

According to particulate hypothesis, parents pass on distinct heritable units. Those distinct heritable units are now called as genes.

RNA purification using glass membrane column?

- ☐ Step 1: Lyse cells, and spin to remove large particulates/cell debris.
- ☐ Step 2: Apply lysate (containing nucleic acids and cellular contaminants) to column with glass membrane.
- ☐ Step 3: Wash with alcohol to remove contaminants; nucleic acids stick to glass membrane while contaminants wash off.
- ☐ Step 4: Treat with DNase enzymes to remove contaminating DNA.
- ☐ Step 5: Apply water to the column; purified RNA washes off the glass membrane and is collected.

Lysis of Leukocytes

RNA purification –blood

Blood cells are lysed in two separate procedures, erythrocytes lysis and leukocytes lysis.

ERYTHROCYTES LYSIS

- ☐ Erythrocytes (red blood cells) do not contain nuclei They neither synthesize nor contain RNA.



Zoology Legends

RNA IS PURIFIED FROM LEUKOCYTES

- ☐ RNA isolation from whole blood is targeted to leukocytes (white blood cells).
- ☐ Leukocytes are nucleated and contain RNA.

ERYTHROCYTES LYSIS BY HYPOTONIC BUFFER

- ☐ Erythrocyte lysis by hypotonic buffer.
- ☐ Intact leukocytes are then recovered by centrifugation. Leukocytes LYSIS
- ☐ Leukocytes lysis by denaturing buffer that immediately, inactivate RNases, allowing isolation of intact RNA.

Effect of environment on allelic expression:

Allele expression is influenced by environmental conditions, as seen in arctic foxes, Himalayan rabbits, and Siamese cats. For instance, the ch allele in Siamese cats is impacted by temperatures above 33°C, causing inactivation of the tyrosinase enzyme, leading to reduced melanin pigment and resulting in their characteristic coat color patterns.

Nonsense mutation & Missense mutation

Nonsense mutation: Nonsense mutations convert an amino acid into a stop codon. The effect is to shorten the resulting protein. Sometimes this has only a little effect, as the ends of proteins are often relatively un-important to function. However, often nonsense mutations result in completely non-functional proteins.

Missense mutation: Missense mutations substitute one amino acid for another. Some missense mutations have very large effects. Some have minimal or no effect.



Zoology Legends

It depends on where the mutation occurs in the protein's structure and how big a change in the type of amino acid it is. Missense mutation produces a change in amino acid sequence in protein product (e.g. Histidine in for Arginine). May change function of protein or may not!

Huntington disease

- ☐ One gene – more than one phenotypes
- ☐ Phenotype 1: progressive dementia at age of 40-50 years
- ☐ Phenotype 2: death in about 5 years after onset of disease.

Epigenetics and fragile X chromosome

Fragile X syndrome is most commonly caused by a CGG trinucleotide repeat expansion in the 5' region of the FMR1 gene. Unaffected individuals have 6-50 CGG repeats. 200 CGG repeats is seen in individuals with fragile X. >200 CGG repeats is correlated with hyper methylation at CpG dinucleotides and silencing.

Use of mgcl2 and EDTA:

Magnesium chloride (MgCl_2) is used in molecular biology reactions, like PCR, as a cofactor for DNA polymerase activity, enhancing enzyme efficiency. Ethylene diamine tetra acetic acid (EDTA) is employed to chelate divalent metal ions like Mg^{2+} , inhibiting DNases and RNases, preserving DNA and RNA integrity.



Zoology Legends

What is the difference between spermatogenesis and oogenesis?

Spermatogenesis:

1. Occurs in the testes.
2. Results in the production of four functional sperm cells.
3. Begins at puberty and continues throughout the male's life.
4. Produces haploid sperm cells.
5. Meiosis divisions are continuous, resulting in minimal cytoplasmic differentiation.
6. Produces millions of sperm cells daily.

Oogenesis:

1. Occurs in the ovaries.
2. Results in the production of one functional egg cell (ovum) and three polar bodies.
3. Begins before birth and ceases during menopause.
4. Produces haploid egg cells.
5. Meiosis divisions are discontinuous, with extensive cytoplasmic differentiation.
6. Produces one egg cell per month (in fertile women).

Karyotyping: We can identify chromosomes through various techniques, such as karyotyping, fluorescent in situ hybridization (FISH), or chromosomal



Zoology Legends

microarray analysis (CMA). These methods allow us to visualize and analyze the structure and number of chromosomes in a cell.

Define chromosomal aberration? How do we classify them (5)

Chromosomal Aberrations: any change in normal number or structure of chromosomes. Affects individuals may be due altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes

Classification

- Numerical (usually due to de novo error in meiosis).
- Structural (may be due to de novo error in meiosis or inherited).
- Different cell lines (occurs post-zygotically).

What is fertilization?

Fertilization is a process of fusion of spermatozoa and ova to produce a cell called zygote.

Diploid - somatic cells

Haploid - sex cells

Monohybrid cross and Dihybrid cross

Cross-fertilization of true-breeding plants which are different in just one character is called as monohybrid cross.



Zoology Legends

With monohybrid cross, Mendel determined the segregation of alleles at single gene locus.

Dihybrid cross determines that alleles at two different gene loci segregate dependently or independently.

Example:

Seed shape is controlled by one gene while seed color is controlled by a different gene. Mendel crossed two pure-breeding plants: one with round/yellow seeds, other with green/wrinkled seeds.

At least three mitochondrial disease

- ☐ Myoclonic epilepsy
- ☐ Mitochondrial recessive ataxia
- ☐ Leber hereditary optic neuropathy
- ☐ Sensory ataxia neuropathy

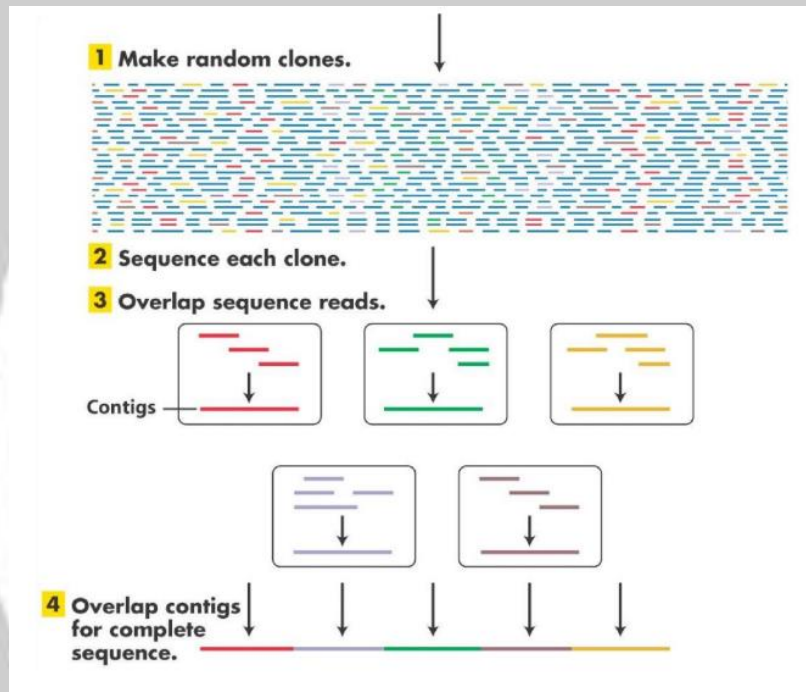
Replication slippage

Replication slippage is an error in DNA replication that can produce duplications of short genetic sequences. During replication DNA polymerase begins to copy the DNA.



Zoology Legends

Bottom up mapping technique



Enlist two epigenetic mechanisms?

- DNA methylation (gene silencing)
- Histone acetylation (gene activation).

Is there any rule for mutation nomenclature?

- ☐ All variants should be described at the most basic level, i.e. DNA level.
- ☐ Descriptions should always be in relation to a reference sequence, either a genomic or a coding DNA sequence.
- ☐ Although theoretically a genomic reference sequence seems best, in practice a coding DNA reference sequence is preferred.



Zoology Legends

- ☐ Good for promoters, alternative splicing etc.
- ☐ When the entire genomic sequence is not known, a cDNA reference sequence should be used.
- ☐ To avoid confusion in the description of a variant it should be preceded by a letter indicating the type of reference sequence used.
- ☐ Several different reference sequences can be used.
- ☐ "c." for a coding DNA sequence (like c.76A>T)
- ☐ "g." for a genomic sequence (like g.476A>T)
- ☐ "m." for a mitochondrial sequence (like m.8993T>C,)
- ☐ "r." for an RNA sequence (like r.76a>u)
- ☐ "p." for a protein sequence (like p.Lys76Asn)
- ☐ Describing genes / proteins, only official HGNC (HUGO gene nomenclature committee) gene symbols should be used (www.genenames.org)
- ☐ DNA reference sequence used should preferably be from the RefSeq database, listing both database accession and version number (like NM_004006.2)

Comparison between Random Amplification Polymorphism DNA or Restriction fragment length Polymorphism?/ RAPD and AFLP

Random Amplified Polymorphic DNA

- ☐ PCR based marker with 10-12 base pairs
- ☐ Random amplification of several fragments



Zoology Legends

- ☐ Amplified fragments run in agarose gel detected by ethidium bromide.
- ☐ Unstable amplification leads to poor repeatability

Amplified Fragment Length Polymorphism

- ☐ Restriction endonuclease digestion of DNA
- ☐ Ligation of adaptors
- ☐ Amplification of ligated fragments
- ☐ Separation of the amplified fragments via electrophoresis and visualization
- ☐ AFLPs have stable amplification and good repeatability

Diagnostic tests for genetic disorders

- **Physical analysis:** Physical analysis for affected individual.
- **Ultrasonography of fetus:** Ultrasonography of fetus determines malformations of head, internal organs, and extremities.
- **Analysis of amniotic fluids:** Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.
- **Maternal blood analysis:** Maternal blood analysis to observe abnormal fetus.

Difference b/w Meiosis I meiosis II

Meiosis I:

- Prophase I: Homologous chromosomes pair up and exchange genetic material (crossing over).



Zoology Legends

- Metaphase I: Tetrads align at the cell's equator.
- Anaphase I: Homologous chromosomes separate.
- Telophase I: Two haploid cells form.

Meiosis II:

- Prophase II: Chromosomes condense in each haploid cell.
- Metaphase II: Chromosomes align at the equator.
- Anaphase II: Sister chromatids separate.
- Telophase II: Four genetically distinct haploid cells form after cytokinesis.

Precautions of RNA purification

- ☐ Immediately vortex and place on ice for 10 minutes.
- ☐ Samples are centrifuged at 14,000 -17,000 rpm for 10 minutes at 4°C.
- ☐ Homogenization, phase separation and RNA precipitation are the steps once pellets of WBC were prepared.

Why polymorphism study in organisms?

Essential to study inheritance patterns. Map phenotypes and anchor genes to the genetic map by cosegregation analysis. Determine change in function: resistant/sensitive populations genetically determined variability among humans is due to a difference in 0.1% of the genomic sequence.



Zoology Legends

Nucleotides, nucleoside, purines, pyridine define

NUCLEOTIDE

Made of three components 5 carbon sugar (pentose) nitrogenous base phosphate group

- Oligonucleotide – short polymer (<10)
- Polynucleotide – long polymer (>10)

Nucleoside = monomer of sugar + base

Nitrogenous Bases pyrimidines consists of cytosine, thymine, uracil

Purines consists of Adenine and Guanine

Turner syndrome

Turner syndrome is a genetic disorder that affects females, typically resulting from a missing or partially missing X chromosome. It can lead to various physical and developmental features, including short stature, infertility, heart and kidney issues, and certain learning difficulties

Exceptions to Mendelian inheritance

- Incomplete dominance
- Codominance
- Multiple alleles
- Polygenic traits
- Epistasis
- Pleiotropy



Zoology Legends

- Environmental effects on gene expression
- Linkage
- Sex linkage

Loss of function, Gain of function

Loss of function: Complete or partial loss of the normal function. Structure of protein is so altered that it is no longer works correctly. Mutation can occur in regulatory regions that affect transcription, translation or splicing of the protein. Frequently recessive.

Gain of function: Produces an entirely new trait. Such a mutation causes a trait to appear in inappropriate tissues or at inappropriate times in development. It is frequently dominant.

Try crossing a brown pepper (RRyy) with a yellow pepper (rrYY). Which trait will your offspring (F1 generation) produce?

What traits are produced when you cross two of the peppers found in the F1 generation?

Brown Pepper RRyy, Yellow pepper rr YY

Spontaneous mutation and it's mutagens

Spontaneous mutation: Mutations that result from natural changes in DNA

Induced mutation results from changes caused by environmental, chemicals and radiations.



Zoology Legends

Mutagen: Any environmental agent that increases the rate of mutation above the spontaneous is called a mutagen such as chemicals & radiations, X-rays, Mustard gas etc

Negative and positive epistasis

Positive epistasis – double mutation

- Double mutation has a fitter phenotype than either single mutation.
- Positive epistasis between beneficial mutations generates improvements in function.

Negative epistasis – two mutations have smaller effect

- When two mutations together have a smaller effect than expected from their effects when alone, it is called negative epistasis.

Interphase

Interphase is a series of changes that takes place in a newly formed cell. Before a cell enters into cell division, it needs to take nutrients.

Interphase proceeds in three stages;

- G1 stage
- S stage
- G2 stage



Zoology Legends

Interphase ~ 90% of the time.

- G1: new cell absorbs nutrients and grows larger.
- S phase: Synthesis of new DNA for daughter cells.
- G2: Cell continues to grow, gets too large, and needs to divide.

Muhammad Zaman

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Remember me in you Precious Prayer

Jazak Allah

ZOOLOGY
LEGENDS